

JEE MAIN REVISION MASTER: SOLID STATE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 23 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Solid State — Core Principles & Governing Laws	13 PYQs	Core Principle
Moderate	Solid State — Applied Cases, Reactions & Traps	6 PYQs	Applied Topics
Moderate	Solid State — Standard Calculations & Problem Archetypes	5 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Solid State — Core Principles & Governing Laws	m^3 , $\rho = 1000 \text{ kg m}^{-3}$, PCL_5
Solid State — Applied Cases, Reactions & Traps	A_7B_{24} , $a=b \neq c$, $\frac{1}{4}$
Solid State — Standard Calculations & Problem Archetypes	$a=300 \text{ pm}$, A_7B_{24} , $r=\frac{a\sqrt{3}}{4}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Solid State — Core Principles & Governing Laws (High Yield — 13 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Solid State.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Solid State.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Solid State PYQs.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): "The number of atoms in 4.5 g of a face-centred cubic crystal with edge length 300 pm is (Given : Density = 10 g cm^{-3} and $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$)." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "A metal crystallises in a body centred cubic lattice with the metallic radius $\sqrt{3}r(A)$. The volume of the unit cell in m^3 ..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "In solid state, PCL_5 is a/an..." [Ans: N/A]

3.2 Solid State — Applied Cases, Reactions & Traps (Moderate Yield — 6 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Solid State.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which of the following crystals has the unit cell such that $a=b \neq c$ and $\alpha=\beta=90^\circ$, $\gamma=120^\circ$?..." [Ans: N/A]
- [2021 Shift] Q#3 (MCQ): "In chrysoberyl, a compound containing beryllium, aluminium and oxygen, oxide ions form cubic close packed structure. Aluminium ions occupy $\frac{1}{2}$..." [Ans: N/A]
- [2021 Shift] Q#4 (MCQ): "The correct statement regarding defects in solid is..." [Ans: N/A]

3.3 Solid State — Standard Calculations & Problem Archetypes (Moderate Yield — 5 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Solid State.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Shift] Q#3 (MCQ): "If 'a' stands for the edge length of the cubic systems. The ratio of radii in simple cubic, body centred cubic and face centred cubic un..." [Ans: N/A]
- [2022 Shift] Q#1 (MCQ): "Which of the following is not true about the amorphous solids?..." [Ans: C]
- [2022 Shift] Q#3 (MCQ): "Vacant space in body centered cubic lattice unit cell is about..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#2)

The number of atoms in 4.5 g of a face-centred cubic crystal with edge length 300 pm is (Given : Density = 10 g cm^{-3} and $N_A = 6.022 \times 10^{23}$)

- (A) 6.6×10^{20}
- (B) 6.6×10^{23}
- (C) 6.6×10^{19}
- (D) 6.6×10^{22}

Correct Answer: None

Solution Analysis: Given, Face centred crystal, so $Z=4$ edge length $(a)=300 \text{ pm}=300 \times 10^{-10} \text{ cm}$ density $(d)=10 \text{ g cm}^{-3}$
 $N_A=6.022 \times 10^{23}$ Using the formula $d = \frac{Z \times M}{N_A \times a^3}$ $M = \frac{d \times N_A \times a^3}{Z} = \frac{10 \times 6.022 \times 10^{23} \times (300 \times 10^{-10})^3}{4} \Rightarrow M = 40.65 \text{ g}$ So, 40.65 g contains 6.022×10^{23} atoms Let 4.5 g contains x atom $\Rightarrow x = \frac{6.022 \times 10^{23} \times 4.5}{40.65} = 6.6 \times 10^{22}$ atoms

Benchmark Problem 2: KCET 2023 (Q#3)

If 'a' stands for the edge length of the cubic systems. The ratio of radii in simple cubic, body centred cubic and face centred cubic unit cells is

- (A) $1a : \sqrt{3}a : \sqrt{2}a$
- (B) $(1/2)a : \frac{\sqrt{3}}{4}a : \frac{1}{2}\sqrt{2}a$
- (C) $(1/2)a : \frac{\sqrt{3}}{2}a : \frac{\sqrt{2}}{2}a$
- (D) $(1/2)a : \sqrt{3}a : \frac{1}{2}\sqrt{2}a$

Correct Answer: None

Solution Analysis: For simple cube, $r = (a/2)$ For bcc, $r = \frac{\sqrt{3}}{4}a$ For fcc, $r = \frac{\sqrt{2}}{4}a$ where, a = edge length, r = radius
Thus, ratio of radii of the three unit cells will be $(1/2)a : \frac{\sqrt{3}}{4}a : \frac{1}{2}\sqrt{2}a$

Benchmark Problem 3: KCET 2021 (Q#3)

In chrysoberyl, a compound containing beryllium, aluminium and oxygen, oxide ions form cubic close packed structure. Aluminium ions occupy $(1/4)$ th of octahedral voids. The formula of the compound is

- (A) BeAlO_4
- (B) BeAl_2O_4
- (C) Be_2AlO_2
- (D) BeAlO_2

Correct Answer: None

Solution Analysis: Given, chrysoberyl forms ccp structure. Oxygen occupies lattice points = N (say) Al occupies $(1/2)$ of octahedral voids = $(N/2)$ Be occupies $(1/8)$ of tetrahedral voids = $(2N/8)$ $\therefore$ The formula of compound is $\text{Be}:\text{Al}:\text{O} (2N/8):(N/2):N \Rightarrow \text{BeAl}_2\text{O}_4$